Predicting ED-to-Ward Waiting Times in Singapore Public Hospitals

Group 6_P2

Colab Notebook:
https://colab.research.google.com/drive/1nO0MMx0arjd01Pf04JKlX4C9bblVsWzY?usp=sharing

Github:
https://github.com/Corv123/Machine-Learning-Proj

Business Understanding

Problem Objective

- Predict median waiting time (hours) for admission from Emergency Department to hospital ward
- Enable hospitals to forecast patient flow and optimise resource allocation

Decision Context

Stakeholder

- Hospital operations: Staff scheduling, bed management
- Patients: Expectation setting, hospital selection
- MOH: Resource allocation across public hospitals

Success Criteria

- Primary: MAE < 3 hours (beat naive mean baseline)
- Secondary: Model explains meaningful variance (R^2 > 0.3)
- Operational: Predictions within realistic range

Data Understanding

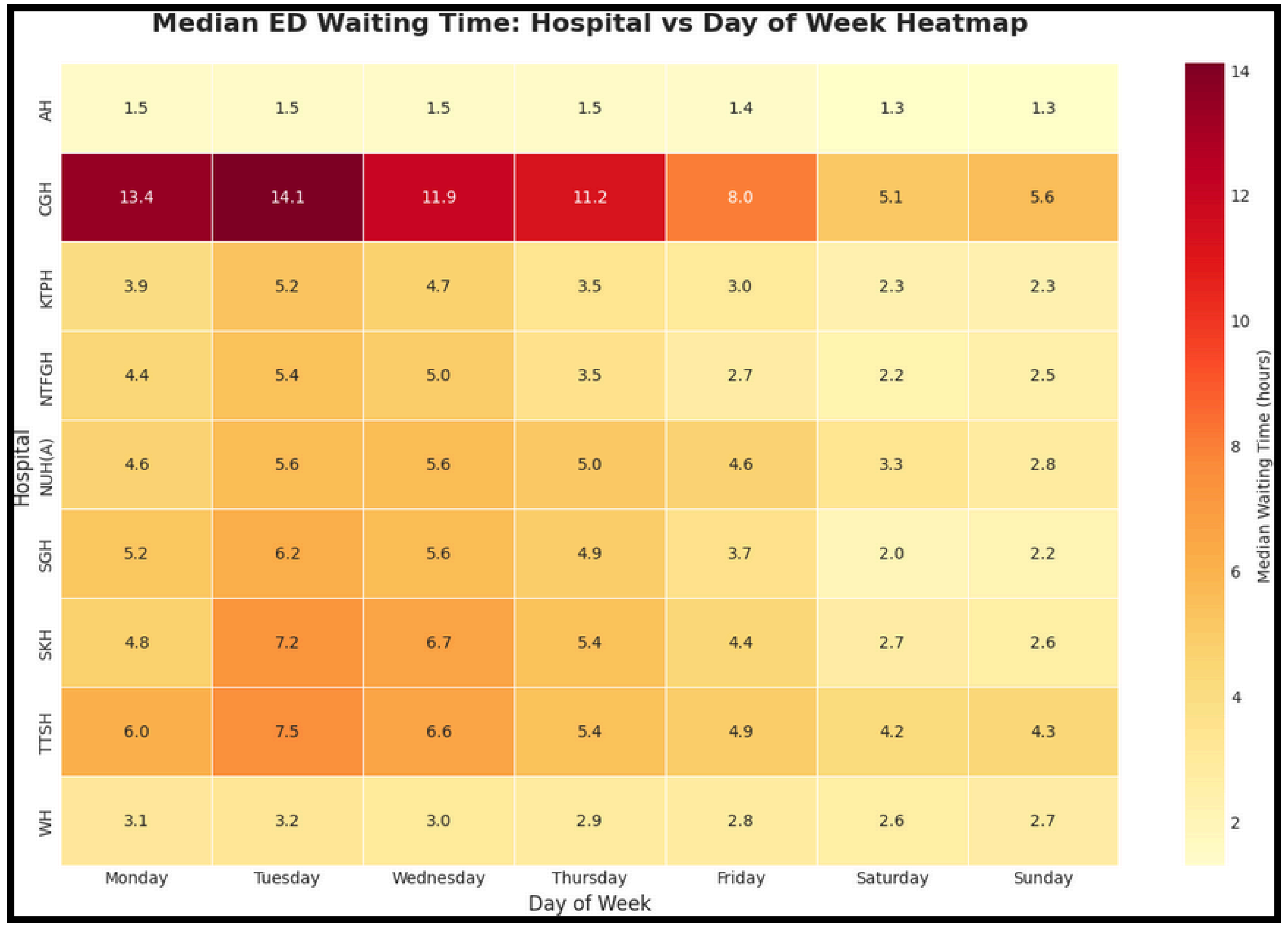
Target Variable

- Waiting time: Median hours from ED arrival to ward admission

Final Dataset

- 9,415 rows x 7 columns (after cleaning)
- 9 Hospitals: Singapore Hospital, Alexandra Hospital, Tan Tock Seng Hospital etc.

Dataset	Granularity	Period
Waiting time for admission to ward	Daily x Hospital	Jan2023 - Jan2026
ED Attendance	Daily x Hospital	Jan2023 - Jan2026
Bed Occupancy Rate	Daily x Hospital	Jan2023 - Jan2026
Public Holidays	Date List	2023 - 2026



Date	Day	Hospital	Waiting Time	Attendance	BOR	Is_Holiday
1/1/2023	Sunday	AH	1.6	64	80.5	1
1/1/2023	Sunday	CGH	7.7	351	89.6	1
1/1/2023	Sunday	KTPH	2.9	286	96.4	1
1/1/2023	Sunday	NTFGH	7.9	252	91.1	1
1/1/2023	Sunday	NUH(A)	2.1	257	78.3	1
1/1/2023	Sunday	SGH	1.3	309	79.3	1
1/1/2023	Sunday	SKH	3	333	86	1
1/1/2023	Sunday	TTSH	3.8	336	91.2	1
2/1/2023	Monday	AH	2	61	84.8	0
2/1/2023	Monday	CGH	12.7	386	90.6	0
2/1/2023	Monday	KTPH	12.2	326	97.9	0
2/1/2023	Monday	NTFGH	4.2	314	92	0

Data Understanding

Waiting Times (50th Percentile) for Admission from ED									
Date	AH	CGH	KTPH	NTFGH	NUH(A)	SGH	SKH	TTSH	WH
Sun, 01/01/23	1.6	7.7	2.9	7.9	2.1	1.3	3.0	3.8	
Mon, 02/01/23	2.0	12.7	12.2	4.2	2.5	2.5	2.7	[illegible]	
Tue, 03/01/23	2.1	16.0	12.9	22.8	4.0	9.3	5.0	4.9	
Wed, 04/01/23	2.5	16.3	21.4	14.7	8.2	11.6	10.3	6.9	
Thu, 05/01/23	2.4	21.5	20.7	18.0	5.2	11.7	7.9	5.0	
Fri, 06/01/23	2.8	18.3	20.0	8.2	6.4	6.8	4.8	4.9	
Sat, 07/01/23	1.3	11.5	6.8	2.4	3.7	1.7	7.7	5.1	
Sun, 08/01/23	1.2	5.4	2.8	2.6	6.1	1.7	5.0	4.8	
Mon, 09/01/23	3.3	11.1	17.9	6.7	5.4	7.6	13.9	5.4	
Tue, 10/01/23	9.0	17.8	6.2	9.7	5.3	6.6	15.0	[illegible]	
Wed, 11/01/23	2.2	18.3	6.7	12.0	4.8	6.1	16.3	6.8	
Thu, 12/01/23	2.4	14.0	11.4	9.5	5.4	3.1	5.6	4.6	
Fri, 13/01/23	1.4	7.9	3.5	5.5	3.8	2.0	8.3	5.2	
Sat, 14/01/23	1.2	3.6	4.8	2.5	2.3	0.8	5.2	4.1	

Issue: Missing values

Finding: Woodlands Hospital data unavailable before June 2024

Action: Dropped

Fri, 18/04/25		0.4	4.6	2.8	2.7	3.3	1.5
Mon, 22/12/25	1.7	21.5	3.3	2.9	4.6	6.4	5.2
Mon, 27/03/23	3.1	12.9	40				

Issue: Outliers

Finding: Waiting time ranges 0.4 - 40.0hrs+

Action: Retained (Reflects real variance → Admitted into Emergency department does not equal to admitted in a ward)

Assumptions

1. Waiting time is accurately recorded at 50th percentile (median)
2. Hospital operational patterns remained relatively stable
3. Historical patterns are predictive of future waiting times

Risks

1. Temporal leakage: Used only same-day features; no future data
2. Distribution shift: Model may degrade if policies/ capacities change

Data Preparation

Cleaning steps:

1. Parsed date formats across three source files
2. Aligned to common date range (Jan 2023 - Jan 2026)
3. Merged on Date using inner join
4. Dropped 539 rows (5.4%) with missing values (WH pre-June 2024)

```
Common starting date (latest of the three starts)
mmon_start = max(wt["Date"].min(), emd["Date"].min(), bor["Date"].min()
int("Common starting date:", common_start.date())
```

```
inner merge on Date so the final dataset only contains dates present in all three
bined = wt_f.merge(emd_f, on="Date", how="inner").merge(bor_f, on="Date", how="inne
bined = combined.sort_values("Date").reset_index(drop=True)
```

```
Drop rows with missing values (WH before June 2024)
nal_df = final_df.dropna(subset=['Waiting Time', 'Attendance', 'BOR']
```

Encoding

Variable	Type	Encoding
Hospital	Categorical (9 levels)	One-hot (drop_first=True)
Day	Categorical (7 levels)	One-hot (drop_first=True)
Is_Holiday	Binary	Used as-is
Attendance, BOR	Numeric	Used-as-is

Train-Test Split

- 80% train / 20% test
- Random state = 42 for reproducibility

Baseline Modelling

Dummy Baseline

- DummyRegressor (mean strategy)
- Estabilishes minimum performance threshold

Classical Models (Default parameters)

- Linear Regression
  → Assumes linear relationships
  → Fast and interpretable
- Decision Tree Regressor
  → Captures non-linear patterns
  → Models feature interactions

Features

- Hospital, Day, Attendance, BOR, is_Holiday

Baseline Results

Performance Comparison

Model	MAE	RMSE	R^2
Dummy(mean)	3.358706	4.741237	-0.000571
Linear Regression	2.439310	3.666226	0.401722
Decision Tree	2.825199	5.002954	-0.114083

Key Findings

- Linear Regression outperforms other models
- 27.4% MAE improvement over dummy baseline
- MAE of 2.44 hours = typical prediction error
- R2 of 0.40 = explains 40% of variance

Critical Insight

- Decision Tree has negative R2 (0.11)
- Performs worse than dummy baseline
- Indicates overfitting with default params
- Strong justification for Stage 2 tuning

Initial Evaluation

Sanity Checks

1) Linear regression had 27.4% improvement over baseline.

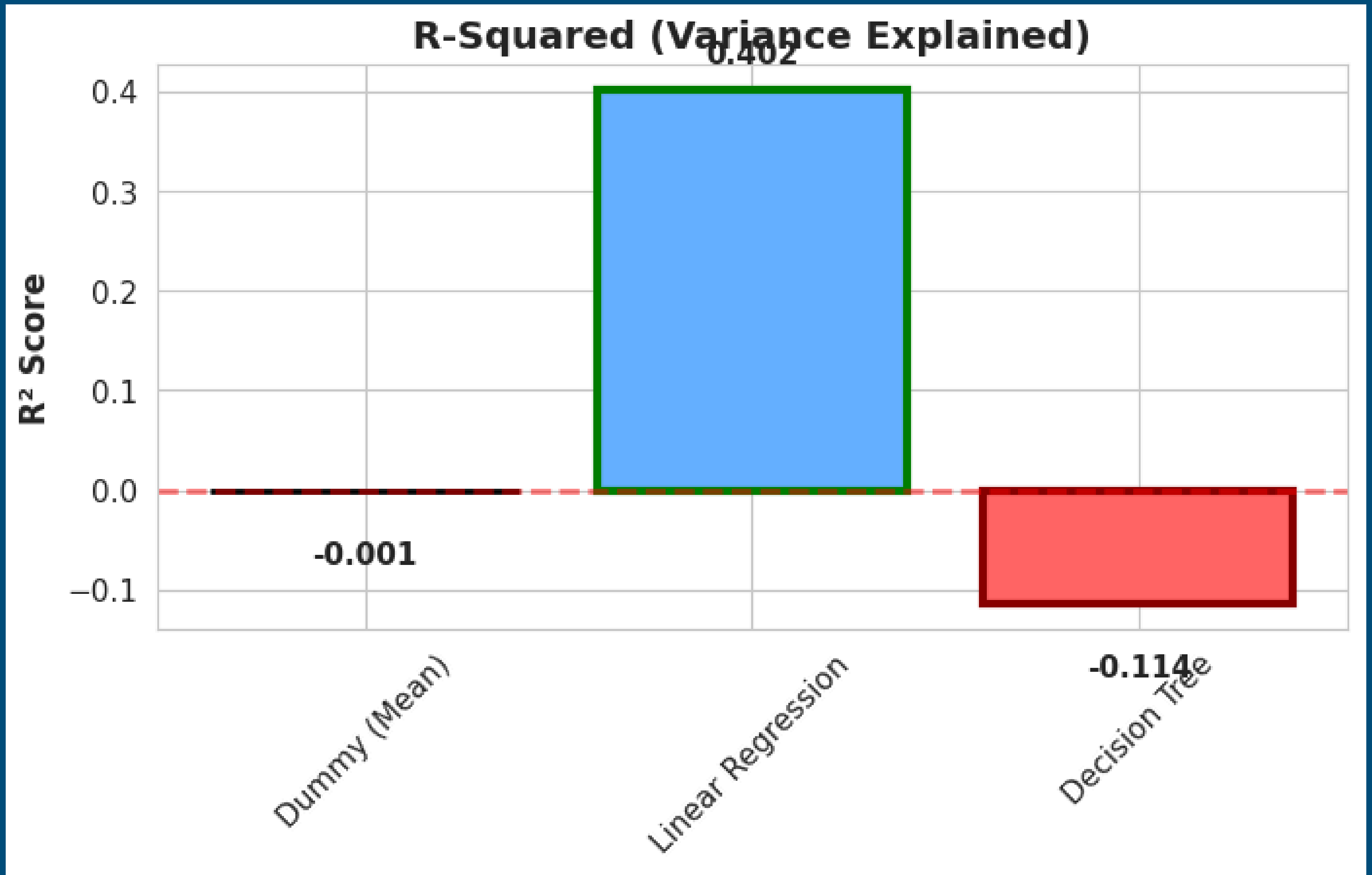
2) Predictions within realistic range

Why Did Decision Tree Fail?

- No max_depth limit = overfitting
- Memorized training data, can't generalize
- Needs hyperparameter tuning in Stage 2

Key Insights

- Linear relationships work well for this data
- 40% of variance explained (R^2 = 0.40)
- Average prediction error: 2.4 hours
- 60% variance unexplained → room for improvement




```
================================================================
SANITY CHECK 3: Feature Importance (Linear Regression)
================================================================
Top 10 features by absolute coefficient:
       Feature  Coefficient
  Hospital_TTSH    -3.363767
 Hospital_NTFGH    -2.843702
   Hospital_SGH    -2.594089
   Hospital_SKH    -2.049186
   Hospital_CGH     1.942564
Hospital_NUH(A)    -1.823283
  Hospital_KTPH    -1.649992
    Day_Tuesday     1.365899
    Hospital_WH    -1.317513
     Day_Sunday    -1.312802
```

Next Steps

Modelling Improvements

Action	Rationale
Tune Decision Tree (max_depth, min_samples, leaf)	Fix overfitting observed in baseline (negative R^2)
Try Random Forest	Ensemble of trees for better generalisation
Ridge/Lasso Regression	Handle multicollinearity from one-hot encoding

Feature Engineering

Feature	Description
Lagged waiting time	Previous 1,3,7-day averages
Delta Occupancy	Change in BOR vs 48-hr average
Rolling ED attendance	3-day moving average of ED visits

External Data Sources

Data	Rationale
Infectious disease rates (flu/COVID)	More actionable for hospital managers
Air quality (PSI/PM2.5)	Haze periods drive respiratory admissions in Singapore

Alternative Problem Framing

Feature	Description
Multi-class classification	Predict Low/Medium/High wait hours

References

References

- MOH : https://www.moh.gov.sg/others/resources-and-statistics/
- Singapore Public Holidays: data.gov.sg

Tools Used

- Google Colab

Links

- Colab Notebook: https://colab.research.google.com/drive/1nO0MMx0arjd01Pf04JKlX4C9bblVsWzY?usp=sharing
- Github: https://github.com/Corv123/Machine-Learning-Proj

Individual Contributions

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